

A02 - Deterministic Fault Campaign

Deterministic fault-injection campaign covering Host timeout, controller-side NAND failures, data-integrity mismatch detection, firmware-log correlation, reset, retry, and readback validation.

STOPPED

9 TOTAL	6 EXECUTED	6 PASSED	0 FAILED
0 ERROR	0 ABORTED	3 NOT RUN	6.861 s DURATION

Suite ID	A02
Suite Name	Automated Fault Injection Campaign
Generated	2026-09-22 01:24:04
Fault Profiles	4
Storage Reset at Start	true
Storage Reset After Campaign	false
Stop Reason	User requested stop
Execution State	6 executed / 3 not run

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Campaign Matrix

Case	Fault	Cmd	LBA	Expected Behavior	Recovery	Result
A02-FP01-C01	timeout	READ	0	Host TIMEOUT -> Reset -> Retry READ	Reset + Retry	PASS
A02-FP01-C02	timeout	READ	100	Host TIMEOUT -> Reset -> Retry READ	Reset + Retry	PASS
A02-FP01-C03	timeout	WRITE	200	Host TIMEOUT -> Reset -> Retry WRITE -> Readback	Reset + Retry + Readback	PASS
A02-FP02-C01	NAND_READ_FAIL	READ	300	FAILED CQ + NAND_READ_FAIL + FW log correlation	Log correlation / No retry	PASS
A02-FP02-C02	NAND_READ_FAIL	READ	1500	FAILED CQ + NAND_READ_FAIL + FW log correlation	Log correlation / No retry	PASS
A02-FP03-C01	NAND_PROGRAM_FAIL	WRITE	400	FAILED CQ + storage unchanged + FW log correlation -> Reset -> Retry WRITE -> Readback	Correlate + Reset + Retry + Readback	PASS
A02-FP03-C02	NAND_PROGRAM_FAIL	WRITE	2000	FAILED CQ + storage unchanged + FW log correlation -> Reset -> Retry WRITE -> Readback	Correlate + Reset + Retry + Readback	NOT_RUN
A02-FP04-C01	miscompare	READ	500	CQ SUCCESS + data mismatch detected; storage unchanged	None	NOT_RUN
A02-FP04-C02	miscompare	READ	2500	CQ SUCCESS + data mismatch detected; storage unchanged	None	NOT_RUN

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Fault Coverage Summary

Profile	Fault Type	Total	Exec	Pass	Fail	Err	Abort	Not Run	Validation Focus
FP01	Command Timeout	3	3	3	0	0	0	0	Host-side timeout detection followed by Controller Reset and retry with preserved CID progression.
FP02	NAND Read Failure	2	2	2	0	0	0	0	Controller returns FAILED / NAND_READ_FAIL and runtime firmware log is correlated to the command.
FP03	NAND Program Failure	2	1	1	0	0	0	1	Controller returns FAILED / NAND_PROGRAM_FAIL, failed write does not modify storage, then reset / retry / readback recovers.
FP04	Data Miscompare	2	0	0	0	0	0	2	CQ remains SUCCESS while returned data is corrupted; the validator detects the mismatch and storage remains intact.

03

Campaign Case Evidence

FP01 - Command Timeout

Host-side timeout detection followed by Controller Reset and retry with preserved CID progression.

A02-FP01-C01 READ LBA 0	PASS
Fault	timeout
Expected Detection	TIMEOUT
Recovery	Reset + Retry
Validation Checks	16 / 16 PASS
Duration	2.131 s
Fault Result	CID 1 / TIMEOUT / Outstanding [1]
Queue After Timeout	SQ 1/1, CQ 0/0
After Reset	Outstanding [] / next CID 2
Retry READ	CID 2 / SUCCESS / pattern=None / unwritten=True

A02-FP01-C02 READ LBA 100	PASS
Fault	timeout
Expected Detection	TIMEOUT
Recovery	Reset + Retry
Validation Checks	16 / 16 PASS
Duration	2.129 s
Fault Result	CID 1 / TIMEOUT / Outstanding [1]
Queue After Timeout	SQ 1/1, CQ 0/0
After Reset	Outstanding [] / next CID 2
Retry READ	CID 2 / SUCCESS / pattern=TEST_PATTERN_A / unwritten=False

A02-FP01-C03 WRITE LBA 200	PASS
Fault	timeout
Expected Detection	TIMEOUT
Recovery	Reset + Retry + Readback
Validation Checks	18 / 18 PASS
Duration	2.182 s
Fault Result	CID 1 / TIMEOUT / Outstanding [1]
Queue After Timeout	SQ 1/1, CQ 0/0
After Reset	Outstanding [] / next CID 2
Retry WRITE	CID 2 / SUCCESS
Recovery Readback	CID 3 / NEW_PATTERN

FP02 - NAND Read Failure

Controller returns FAILED / NAND_READ_FAIL and runtime firmware log is correlated to the command.

A02-FP02-C01 READ LBA 300	PASS
Fault	NAND_READ_FAIL
Expected Detection	FAILED_CQ_AND_LOG
Recovery	Log correlation / No retry
Validation Checks	14 / 14 PASS
Duration	0.104 s
CQ Completion	FAILED / NAND_READ_FAIL
Firmware Log	CID 1 / READ / LBA 300 / NAND_READ_FAIL
Correlation	CID + Opcode + LBA + Status + Error matched

A02-FP02-C02 READ LBA 1500	PASS
Fault	NAND_READ_FAIL
Expected Detection	FAILED_CQ_AND_LOG
Recovery	Log correlation / No retry
Validation Checks	14 / 14 PASS
Duration	0.104 s
CQ Completion	FAILED / NAND_READ_FAIL
Firmware Log	CID 1 / READ / LBA 1500 / NAND_READ_FAIL
Correlation	CID + Opcode + LBA + Status + Error matched

FP03 - NAND Program Failure

Controller returns FAILED / NAND_PROGRAM_FAIL, failed write does not modify storage, then reset / retry / readback recovers.

A02-FP03-C01 WRITE LBA 400	PASS
Fault	NAND_PROGRAM_FAIL
Expected Detection	FAILED_CQ_AND_LOG
Recovery	Correlate + Reset + Retry + Readback
Validation Checks	22 / 22 PASS
Duration	0.209 s
Fault Completion	FAILED / NAND_PROGRAM_FAIL
Firmware Log	CID 1 / WRITE / LBA 400 / NAND_PROGRAM_FAIL
After Reset	next CID 2
Retry WRITE	CID 2 / SUCCESS
Readback	CID 3 / NEW_PATTERN

A02-FP03-C02 WRITE LBA 2000	NOT_RUN
Fault	NAND_PROGRAM_FAIL
Expected Detection	FAILED_CQ_AND_LOG
Recovery	Correlate + Reset + Retry + Readback
Validation Checks	Not executed
Duration	0.000 s
Execution State	Not run because campaign stop was requested

FP04 - Data Miscompare

CQ remains SUCCESS while returned data is corrupted; the validator detects the mismatch and storage remains intact.

A02-FP04-C01 READ LBA 500	NOT_RUN
Fault	miscompare
Expected Detection	DATA_MISCOMPARE
Recovery	None
Validation Checks	Not executed
Duration	0.000 s
Execution State	Not run because campaign stop was requested

A02-FP04-C02 READ LBA 2500	NOT_RUN
Fault	miscompare
Expected Detection	DATA_MISCOMPARE
Recovery	None
Validation Checks	Not executed
Duration	0.000 s
Execution State	Not run because campaign stop was requested

04

Campaign Conclusions

Campaign Result	STOPPED
Stop Reason	User requested stop
Execution State	6 / 9 cases executed
Aborted Cases	None
Not Run	3
Interpretation	The active Case was cooperatively cancelled when applicable; remaining scheduled Cases were not started and are not treated as failures.

STOP v1: Cancellation is cooperative. Active fake_nvme subprocesses can be terminated and recovered through simulated controller reset; arbitrary Python deadlocks or non-cooperative infinite loops are outside the v1 stop guarantee.

Scope: This is a deterministic Host-side logical NVMe validation campaign. It demonstrates automation, fault detection, recovery reasoning, firmware-log correlation, and data-integrity validation. It does not claim physical SSD firmware or PCIe hardware validation.